# Prediction of co-circulation for multiple respiratory viruses by assessing their dynamical interaction in pre-, during and post-pandemic periods

Dong Wang[1,2], Jiaqi Chen[1], Hao Wang[1], Sukhyun Ryu[3], EHY Lau[3], Peng Wu[1], Daihai He[4], Lulla Opatowski[5], Benjamin J. Cowling[1], Sheikh Taslim Ali[1]

[1] WHO Collaborating Centre for Infectious Disease Epidemiology and Control, School of Public Health, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Hong Kong,China
[2] Research Center for Industries of the Future, Westlake Institute for Advanced Study, Westlake University, Hangzhou, Zhejiang Province, China.
[3] Department of Preventive Medicine, College of Medicine, Konyang University, Daejeon 35365, Korea
[4] Department of Applied Mathematics, The Hong Kong Polytechnic University, Hong Kong, China
[5]Epidemiology and Modelling of Antimicrobials Evasion Research Unit (EMAE), Institut Pasteur, Université de Paris, Paris, France

## Background

Respiratory viruses, including influenza A(H1N1), A(H3N2), influenza B, and SARS-CoV-2 are significantly active in community, may shape one another's dynamics and circulation potential through their respective population immunity and directional effects on infection risk and disease severity. However, the extent to which these interactions are asymmetric and biologically heterogeneous remains unclear.

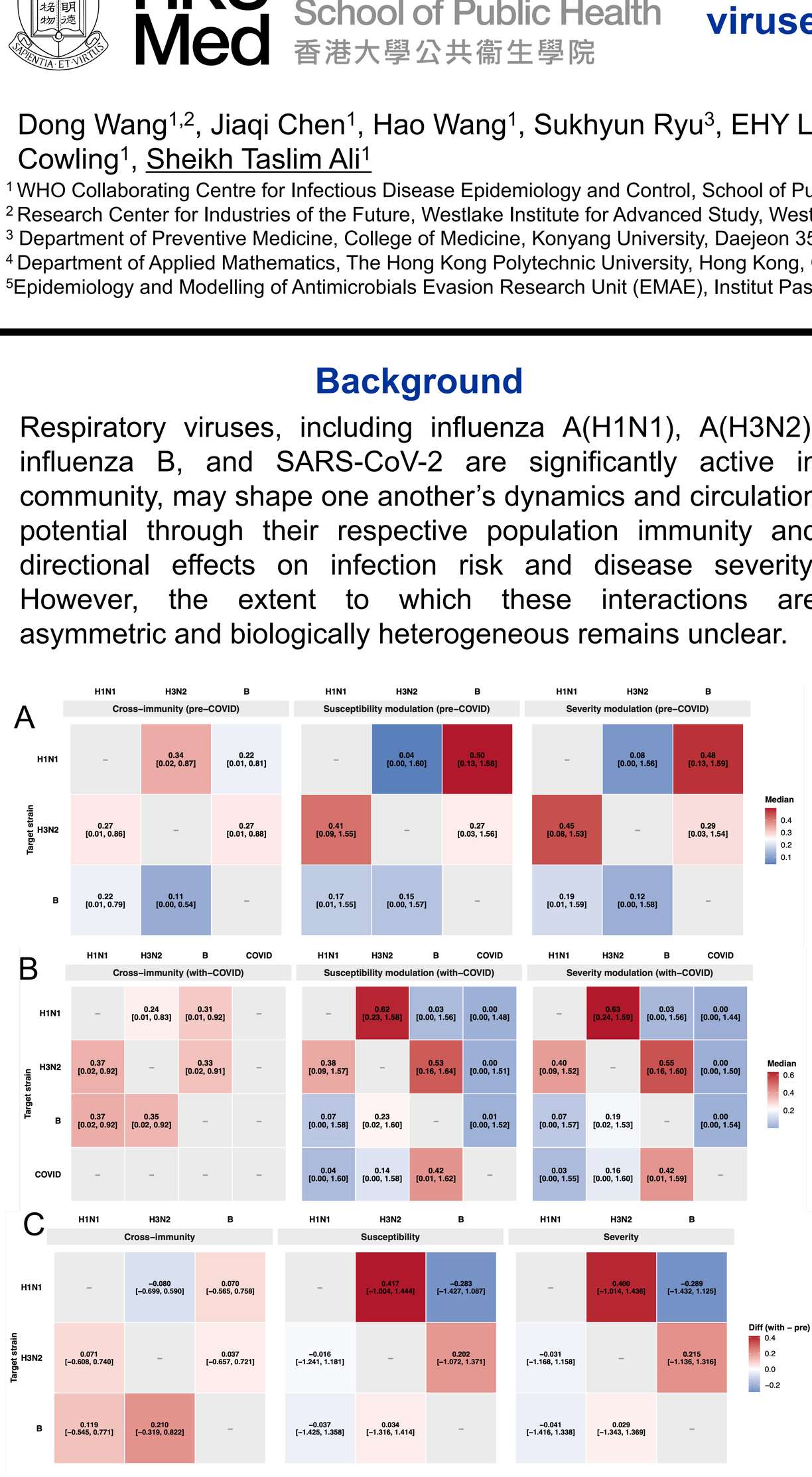



Figure 1. Cross-interaction between influenza subtypes and SARS-CoV-2, posterior median with 95% credible interval (CrI). **Columns represent three interaction mechanisms: Cross-immunity** (infection by the partner strain reduces the target strain's recovered-to-susceptible return rate), **Susceptibility impact** (partner infection modulates the target strain's susceptible pool), and **Severity impact** (partner infection modulates ILI intensity / severity). Rows: (A) Pre-COVID period (3×3, influenza only); (B) With-COVID period (4×4, includes SARS-CoV-2 row/column); (C) Paired per-draw difference (with-COVID minus pre-COVID, 3×3); blue = lower with COVID, red = higher; White: (A,B) global median; (C) zero. Diagonal omitted.

## Discussion

By resolving directional effects among co-circulating viruses, our study suggests that post-pandemic respiratory dynamics are shaped by structured interaction networks rather than independent transmission processes. The particularly distinct profiles observed for influenza B and SARS-CoV-2 highlight their potential roles in driving changes in epidemic composition and timing.

## Methods

We applied state-space multi-strain co-circulation models to weekly surveillance data to estimate directed interactions among influenza A(H1N1), A(H3N2), influenza B, and SARS-CoV-2 before and during the COVID era. Bayesian target-specific model parameterization was used to infer cross-immunity, susceptibility modification, and severity impacts. Posterior medians and 95% credible intervals (CrIs) were used to summarize the magnitude and direction of these interactions.

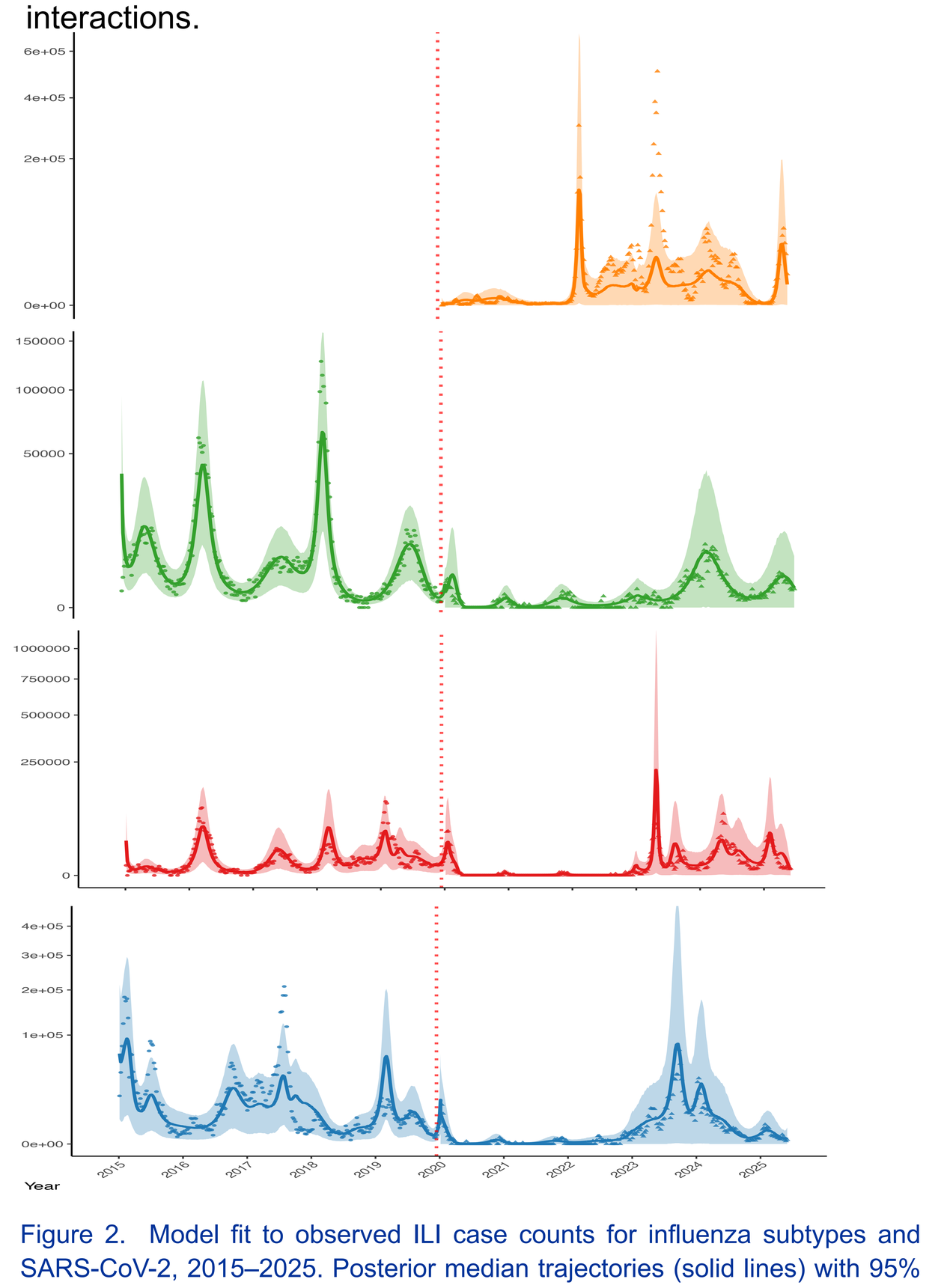



Figure 2. Model fit to observed ILI case counts for influenza subtypes and SARS-CoV-2, 2015–2025. Posterior median trajectories (solid lines) with 95% credible intervals (shaded ribbons) overlaid on weekly observed cases (points). Panels: COVID-19 (orange), Influenza B (green), H1N1 (red), and H3N2 (blue). Circles denote the pre-COVID influenza period (2015–2019); triangles denote the co-circulation period (2020–2025). The vertical dotted line marks the start of 2020. Data: influenza surveillance (2015–2019) and influenza–COVID-19 co-circulation surveillance (2020–2025).

## References

[1] Wong A, Barrero Guevara LA, Goult E, Briga M, Kramer SC, et al. (2023) The interactions of SARS CoV-2 with cocirculating pathogens: Epidemiological implications and current knowledge gaps. PLO Pathogens 19(3): e1011167.

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**Contact: alist15@hku.hk (Presenting Author name)**